Maintained reproduction of Aggarwal, Allen, Coppock, Frankowski, Messing, Zhang, Barnes, Beasley, Hantman and Zheng (2023)

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Archived at Zenodo: doi:10.5281/zenodo.24168257. This DOI always resolves to the latest release of the maintained version.

Drafted by Claude Opus 5 under the supervision of Alex Coppock.

1 What this repository is

This repository re-runs the analysis behind a published article from its deposited data, in current R, and lays every number the article prints against what that analysis produces. It holds three things: a script that fetches and verifies the authors' replication archive, a rewrite of the analysis using current packages, and a ground truth that records, claim by claim, whether the published number reproduces.

Article	Aggarwal M, Allen J, Coppock A, Frankowski D, Messing S, Zhang K, Barnes J, Beasley A, Hantman H and Zheng S (2023). "A 2 million-person, campaign-wide field experiment shows how digital advertising affects voter turnout." *Nature Human Behaviour* 7, 332–341. https://doi.org/10.1038/s41562-022-01487-4
Replication archive	https://doi.org/10.7910/DVN/YMKVA1
Pre-analysis plans	https://osf.io/3evfp and https://osf.io/jkush
Treatment stimuli	https://osf.io/ex3kq

1.1 Folder layout

```
download_original.R    fetches and verifies the deposited archive
run_all.R              runs everything, in order
original/              the deposited archive (not redistributed; fetched on demand)
original_manifest.csv  every deposited file, its size and its published checksum
maintained/            the rewrite; maintained/output/ holds every number it produces
ground_truth/          the extraction, the comparison and the gates
errata.qmd             corrections to the published article
```

1.2 How to reproduce

Download the repository, open `aggarwal_etal_2023.Rproj`, and run:

```
source("run_all.R")
```

`run_all.R` fetches the deposited archive from Harvard Dataverse, verifies every file against its published checksum and byte size, writes every figure and table, computes every in-text quantity, runs the deposited scripts in a scratch copy, rebuilds the ground truth and re-verifies the archive. Nothing writes into `original/`. The archive is 494 MB, most of it one Facebook Ad Library extract, so the first run spends several minutes downloading.

2 The paper

A campaign-level field experiment run by Acronym, a left-leaning non-profit, during the 2020 US presidential election. Of the registered voters in Arizona, Michigan, North Carolina, Pennsylvania and Wisconsin who met the campaign's targeting criteria, a randomly sampled holdout saw none of Acronym's advertising for eight months while everyone else saw the full programme. Turnout was then read from the TargetSmart voter file for the 1,999,282 participants whose assignment could be recovered.

The headline results are an average effect on turnout indistinguishable from zero and a difference in conditional average treatment effects by modelled Trump support of 0.7 percentage points: turnout rose 0.4 points among Biden leaners and fell 0.3 points among Trump leaners.

3 Does the deposited archive run?

The deposit is 16 files: eleven analysis scripts, four data files and a README. Every script was run in a scratch copy, twice, once as the archive ships and once stripped to data plus code. `helper_file.R` is only ever sourced by the other scripts, so it runs inside them and is not counted on its own.

```
Warning in styling_latex_scale(out, table_info, "down"): Longtable cannot be resized.
```

Table 1: Deposited scripts by pass and outcome.

pass	clean	error
as_shipped	4	6
stripped	4	6

The two passes give the same answer, and that is a finding rather than an omission. Stripping the copy to data plus code removes nothing, because no deposited file is written by any deposited script: the write-call scan below returns 3 uncommented writes and all 3 of them target an `output/` subdirectory that is not part of the deposit. The archive ships no derived objects, so the stripped-copy test is vacuous here and is reported as such rather than as a pass.

6 of 10 scripts stop on an error, for four distinct reasons.

```
Warning in styling_latex_scale(out, table_info, "down"): Longtable cannot be resized.
```

Table 2: Where each failing script stopped, as shipped.

script	stopped_at
in_text_calculations.R	Error in waldtest(omni_fit, omni_fit_restricted, test = "...
figure_2.R	Error in xtable(arrange(mutate(select(gg_df, inquiry, out...
figure_3.R	Error in ‘ggsave()‘:
table_1.R	Error in file(file, ifelse(append, "a", "w")) :
figure_B1.R	Error in readRDS(con, refhook = refhook) : cannot open th...
figure_C4.R	Error:

- **Two scripts fail because the deposit does not ship the directory they write into.** `figure_3.R` calls `ggsave("output/figure_3.pdf", ...)` and `table_1.R` opens `output/table_1a.tex` for writing. Neither `output/` nor any `dir.create()` is in the deposit, so on a clean copy both stop at their first write. Every estimate each script computes is already in memory when it stops.
- **Two scripts fail on data files the deposit does not contain.** `figure_B1.R` reads `fielding_dates.rds` and `fielding_dates_survey.rds`; `figure_C4.R` reads `weekly_adds_for_page_id_disclaimer_041322.csv`. The deposit's own README lists all three among the files the appendix scripts depend on. They are the reason Supplementary Figures 1 and 4 have no coverage here.
- **`figure_2.R` fails on a missing package.** It calls `xtable()` after opening with `rm(list = ls())` and without `library(xtable)`, so the call resolves only if a previous script attached the package in the same session.
- **`in_text_calculations.R` fails on a commented-out package.** Its `library(lmtest)` line is commented out and it calls `waldtest()`. Everything above that line runs.

One further defect in the deposit is not visible as a failure, because the script stops before reaching it. Below the `waldtest()` call, `in_text_calculations.R` computes the article's six advertising spending figures with `filter(!is_turnout)` and `group_by(ad_format)` and `group_by(ad_type)`. The deposited `ad_library_2020_acronym.csv` carries none of those three columns; they are built in `figure_B2.R`. So the script that is named for the in-text calculations cannot produce the in-text spending figures even after the `lmtest` line is restored. All six reproduce from `figure_B2.R`'s classification, which is what this rewrite uses.

Running the deposit in place would add 1 file to the archive directory, an `Rplots.pdf` from the scripts that print a plot to the active device, and would overwrite 0 deposited files.

The archive reproduces the article. Every one of the 946 numbers the article's six large floats print is recovered from the deposited code, and none of the 53 ground truth rows with an archive counterpart disagrees with the page.

4 Errata

Seven sentences in the published article are wrong, and they are corrected in `aggarwal_etal_2023_errata.pdf`, whose values are computed at render time from this repository's output. None of them changes a conclusion. In summary: the Supplementary Information names July as the month of heaviest persuasion spending where the figure it points at makes it February, and that cross-reference does not resolve; the Methods count 21 pre-treatment covariates where Figure 5 and Supplementary Table 3 both carry 22; the treatment audience is given as "1,993,216 million" where the sentence's own parenthetical makes it 1,993,216; the balance F test's degrees of freedom are printed as the residual count first and a negative parameter difference second; and one share is rounded down where it rounds up.

4.1 Findings that are not errata

The deposited scripts are not quite the scripts that made the published figures. `figure_3.R` draws two `geom_label` annotations reading "Control group" and "Treatment group"; the published Figure 3 carries neither. `figure_5.R` labels its axis rows "Trump support score / 100", "Turnout score / 100", "Ideology score / 100" and "Partisanship score / 100"; the published Figure 5 prints "TSS/100", "Turnout score/100", "Ideology score/100" and "Partisanship score/100". Every estimate in both figures reproduces exactly. The rewrite follows the deposited code, so its Figure 3 carries the two labels.

One targeting criterion does not hold in the data the article analyses. Figure 4 gives the targeting criteria as a Trump support score between 30 and 70 and a presidential turnout score between 20 and 80. The Trump support score in the deposited analysis set runs exactly 30 to 70. The presidential turnout score runs 20.0 to 99.2, so the upper half of the criterion is not satisfied. The same figure states the archival subset's own criteria as a presidential turnout score above 20 with no upper bound, and the analysis sample is that subset, so the two descriptions in one figure are not consistent with each other. Which is right cannot be settled from the deposit, and no published estimate depends on it, so this is recorded as unresolved rather than corrected.

The state list of the Facebook Ad Library query appears twice and differs. Supplementary Information section C describes the query as covering AZ, MI, NC, PA and WI; the caption of Supplementary Figure 3, which that query produces, lists AZ, GA, MI, NC, PA and WI. The deposited extract carries no state field, so neither list can be checked. The total the section states, US$349,006,000, is the lower-bound sum over the whole deposited extract and reproduces exactly.

The Reporting Summary gives the study's timing as July to November 2020 where the Methods describe an eight-month programme running from March 2020 to election day. The Reporting Summary form does not say which activity it is timing, so this is left as an inconsistency rather than corrected.

The registration date of the pre-analysis plan appears twice and differs. Supplementary Information section D says the plan was registered on 22 November 2020; the plan it reproduces is dated 11 November 2020. A registration can postdate a document, so this is not necessarily an error.

5 The ground truth

`ground_truth/aggarwal_etal_2023_ground_truth.csv` carries one row per published claim, 218 of them. `value_paper` is the string the article prints and is the only column anyone typed; `value_script` is read out of the deposited code and `value_rewrite` out of `maintained/output/`, both by script. A value agrees when the computed number, printed to the page's own precision, gives the same digits.

```
Warning in styling_latex_scale(out, table_info, "down"): Longtable cannot be resized.
```

Table 3: Published claims by type and verdict.

claim_type	does not reproduce	no verdict	reproduces	does not hold	holds
definitional	2	56	13	0	0
descriptive	0	1	0	4	2
pipeline	3	1	88	0	0
structural	0	7	6	3	0
transcribed	0	32	0	0	0

```
Warning in styling_latex_scale(out, table_info, "down"): Longtable cannot be resized.
```

Table 4: Where the fault lies, on every row that is not a clean match.

defect_locus	rows
archive	47
paper_internal	10
unresolved	3

Every adverse row carries a `defect_locus` and no clean match carries one. The five values are `paper_internal` (the article disagrees with its own tables or its own data), `archive` (the deposit cannot support the claim), `environment` (R or a package moved underneath it), `rewrite` (ours) and `unresolved` (cause not established). The `archive` rows here are almost all of one kind: the deposit is the analysis subset and one Ad Library extract, and the article also

describes an assignment population of 31 million voters, a Facebook match rate, an impressions count and a Wesleyan Media Project spending series, none of which the deposit was ever going to carry.

5.1 Coverage, float by float

```
Warning in styling_latex_scale(out, table_info, "down"): Longtable cannot be resized.
```

Table 5: Coverage per published float. Published numbers are what the float prints face, not counting axis breaks; plotted counts what a figure plots, where makes it countable.

float	published numbers	covered	fraction	reproduced by rewrite	reproduced by ar
Fig. 1	114	114	100%	114	
Fig. 2	36	36	100%	36	
Fig. 3	0	0	NA	0	
Fig. 4	27	8	30%	7	
Fig. 5	0	0	NA	0	
Fig. 6	0	0	NA	0	
Table 1	95	95	100%	95	
Supplementary Figure 1	0	0	NA	0	
Supplementary Figure 2	0	0	NA	0	
Supplementary Figure 3	0	0	NA	0	
Supplementary Figure 4	0	0	NA	0	
Supplementary Table 1	399	399	100%	399	
Supplementary Table 2	126	126	100%	126	
Supplementary Table 3	176	176	100%	176	

Six floats carry the bulk of the published record. Supplementary Tables 1, 2 and 3 print 701 numbers between them, main-text Table 1 prints 95, and Figures 1 and 2 print an estimate and a standard error above every point, 114 and 36 numbers, which makes them published tables in disguise. All 946 are parsed from the published PDFs into `ground_truth/published_appendix_values.csv` and `ground_truth/published_maintext_tables.csv` and compared cell by cell. Every one reproduces.

Figures 1 and 2 are read positionally from the text layer, never in reading order, and they are read from the figures themselves rather than from the supplementary tables their captions point at. That makes the agreement between the two published transcriptions a finding: all 114 of Figure 1's cells and all 36 of Figure 2's equal the corresponding supplementary table cell.

Figures 3 and 5 print no numbers on their faces. For those the checkable published quantity is what they plot: Figure 3 plots 246 binned means, which is 41 one-point bins by two conditions by three outcomes, and Figure 5 plots 22 covariate estimates, one per axis label. Both counts are verified and both figures' underlying estimates are written to CSV.

Two floats have zero coverage and the reason is the same in both cases. Supplementary Figure 1 needs `fielding_dates.rds` and `fielding_dates_survey.rds`; Supplementary Figure 4 needs `weekly_adds_for_page_id_disclaimer_041322.csv`. None of the three is among the 16 deposited files, the deposited scripts that read them fail at their first read, and neither figure prints any number other than its axis breaks. Figure 6 is two photographs of advertisements and asserts no quantity at all.

6 The extraction and the two instruments

`ground_truth/published_claims.csv` is the extraction: 218 rows, one per quantitative claim in the article, its figure captions, its Methods, its Reporting Summary and its Supplementary Information, each classified by hand as `pipeline`, `descriptive`, `definitional`, `structural` or `transcribed`, and each carrying the string the page prints and the precision it prints it at.

```
Warning in styling_latex_scale(out, table_info, "down"): Longtable cannot be resized.
```

Table 6: The extraction by claim type and whether the pipeline can reach the quantity.

claim_type	needs a block: FALSE	needs a block: TRUE
definitional	55	16
descriptive	0	7
pipeline	0	92
structural	7	9
transcribed	32	0

It was built by extracting every numeric token from the text layer of both PDFs with its surrounding context, then classifying each by hand. Numbered citation markers, the reference lists, calendar years used only as dates and the article's own DOI are excluded, and the six floats above are carried as one row each with their cells compared separately rather than as 946 rows. A companion pass swept for numbers written as words, which no token scan sees: "five battleground states", "8-month-long", "plus or minus one-third of a percentage point", "only two exhibited statistically significant imbalance" and "greater than three points" all entered the extraction that way.

Two instruments read the same pipeline output by separate paths. `ground_truth/build_ground_truth.R` builds the comparison table. `maintained/in_text_claims.R` carries the article's own sentence in a block comment beside code that recomputes the number, and prints 124 claims in the form `CLAIM <id> = <value> || <label>`. It reads `maintained/output/` and the extraction, never the ground truth, so the two derivations are independent and a disagreement between them is a finding.

The build asserts all of the following and stops if any fails.

- Every `pipeline` and `descriptive` claim has a block and a row.
- Re-rendering `value_paper` from its own number at its own recorded precision returns the string the extraction stores. This runs before anything consumes the precision, because it is the check that catches a precision right about the value and wrong about the digits.
- Where the same published quantity is printed both in prose and in a supplementary table, the two hand transcriptions agree at whichever precision is coarser.
- The number of printed claims equals the number of rows requiring one, and the two sets of ids are equal in both directions.
- Every value the two instruments both produce agrees exactly.
- Every published cell joins to exactly one pipeline value; a published cell that finds nothing stops the build rather than becoming missing.
- Every adverse row carries a `defect_locus` and no clean match does.
- The ground truth and the extraction cover the same claims, in both directions.

Three of these were tested by breaking them, and the checks are ordered so that each break reaches its own: deleting a block fails the count and the id comparison, perturbing a printed value fails the cross-instrument comparison, and changing one claim's recorded precision fails the string check before the value comparison can absorb it.

6.1 What the second instrument found that a float-shaped ground truth would not

Every number in the article's six large floats reproduces, so a ground truth built around floats would have returned a clean sweep. All seven errata, and every finding above, are prose, captions or the document's own structure.

```
Warning in styling_latex_scale(out, table_info, "down"): Longtable cannot be resized.
```

Table 7: Descriptive claims: what the article asserts about shape, sign and count, and whether it holds. The evidence behind each verdict is written out beside it in the pipeline output.

claim	holds

Out of 21 covariates, only two exhibited statistically significant imbalance	FALSE
only two exhibited statistically significant imbalance	TRUE
Neither of these estimates remained significant after a Benjamini-Hochberg correction	TRUE
An F-test ... was non-significant	TRUE
the campaign increased voting among Biden leaners	TRUE
and decreased voting among Trump leaners	TRUE
We also observed small conditional average effect estimates by age, gender, race and vo...	TRUE
we can affirm that our overall estimate is effectively equivalent to zero using the two...	TRUE
differential effects of the programme were stronger in our early voting data than in th...	TRUE
1.0 percentage points favouring Biden ... 0.3 percentage points favouring Trump	TRUE
party registration was not available for a large proportion of the participant pool (72%)	TRUE
we are missing race data for 4% of the sample where race was uncoded	NA
Acronym spent more on persuasion in July than in any other month	FALSE

7 The maintained rewrite

`maintained/` is 12 scripts. Every figure and table script writes the unrounded estimates it plots or prints alongside the figure itself, so nothing downstream reads a number back out of a formatted cell.

The substitutions are the usual ones for code of this age: the native pipe for `magrittr`, `reframe(tidy(...))` for `do(tidy(...))`, `pick(everything())` for the implicit `.` inside `do()`, `if_else()` for `ifelse()`, `separate_wider_delim()` for `separate()`, `knitr::kable()` written to a file for `xtable` printed through a connection, and `label_dollar()`/`label_date()` for the `scales::` formatter aliases. `rm(list = ls())` is dropped, packages are loaded once in `helpers.R`, and every path goes through `here::here()`.

One property of the deposited data has to be handled explicitly. `aggregated_analysis_set.rds` is a grouped tibble, grouped by all 41 of its covariate columns, which is what a `count()` leaves behind. A bare `summarize()` over it returns two million rows rather than one, which is why the deposit's `table_1.R` calls `ungroup()` before its total row. The rewrite drops the grouping when the file is read, where it cannot be forgotten.

The rewrite adds three published tables the deposit computes and never writes. The deposited `figure_1.R`, `figure_2.R` and `figure_5.R` each assemble an `xtable` object holding exactly the cells of Supplementary Tables 1, 2 and 3, and each has its `print.xtable()` call commented out. Those three tables are 701 of the article's published numbers, and without writing them there is no artifact to compare against.

8 Figure verification

Every rendered figure was laid beside the published one. Numbers alone cannot catch a transposed axis or a mislabelled series, and two of this article's four plotted figures print their estimates on the face of the plot.

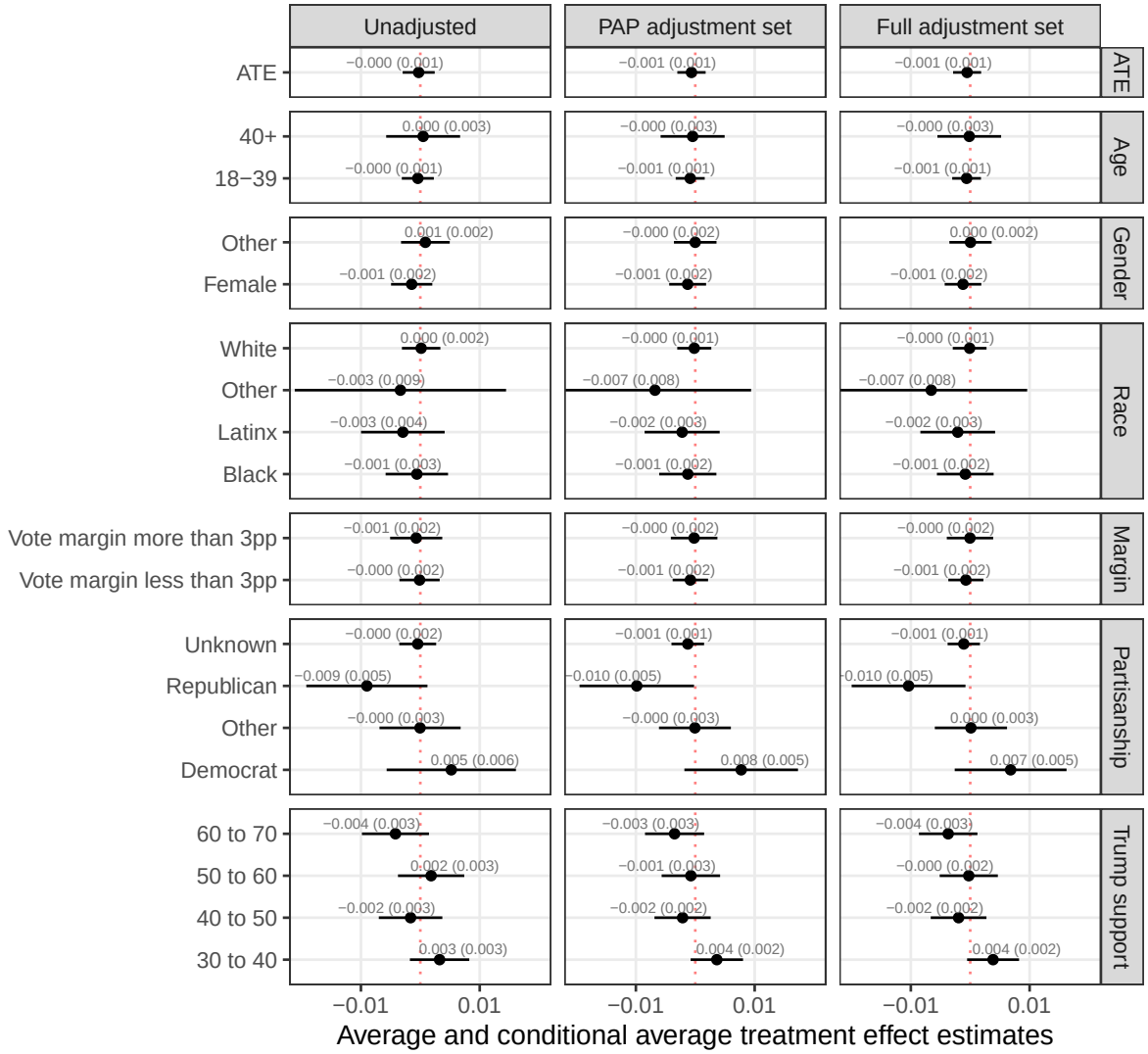


Figure 1: Figure 1 as the rewrite draws it. All 114 printed labels, the facet order and the row order within each facet reproduce the published figure.

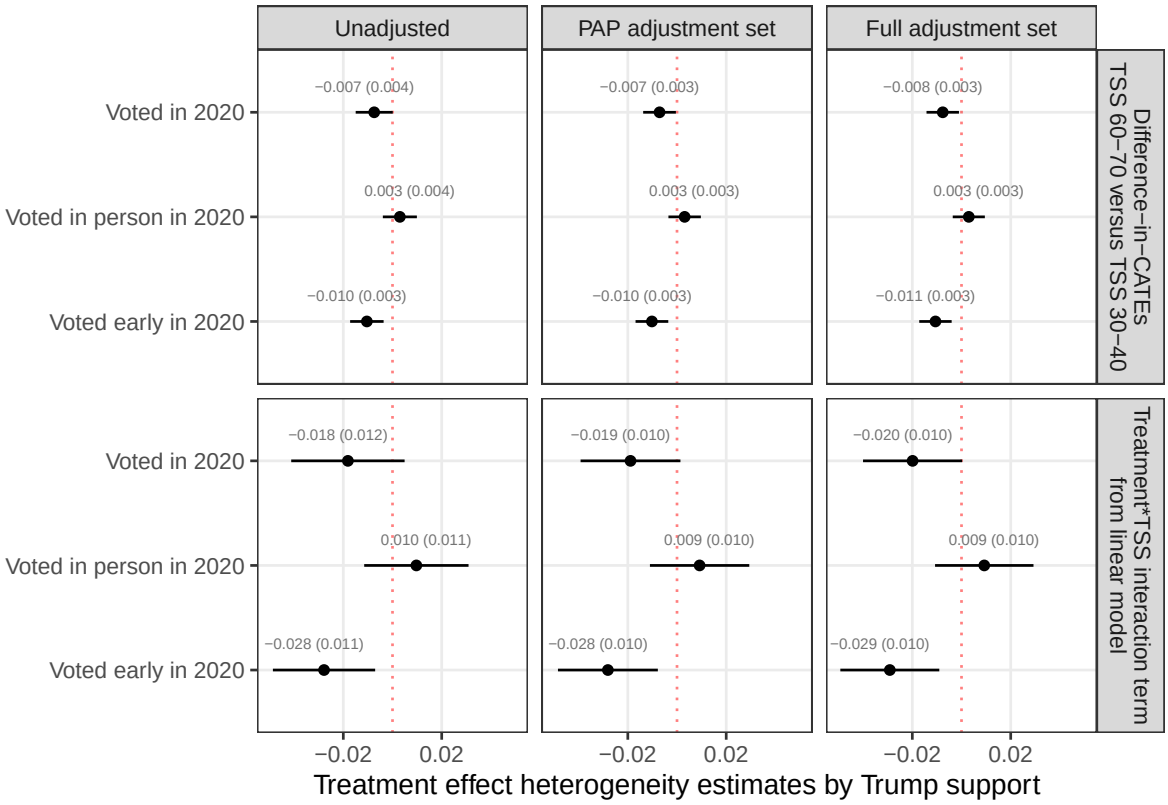


Figure 2: Figure 2 as the rewrite draws it. All 36 printed labels reproduce.

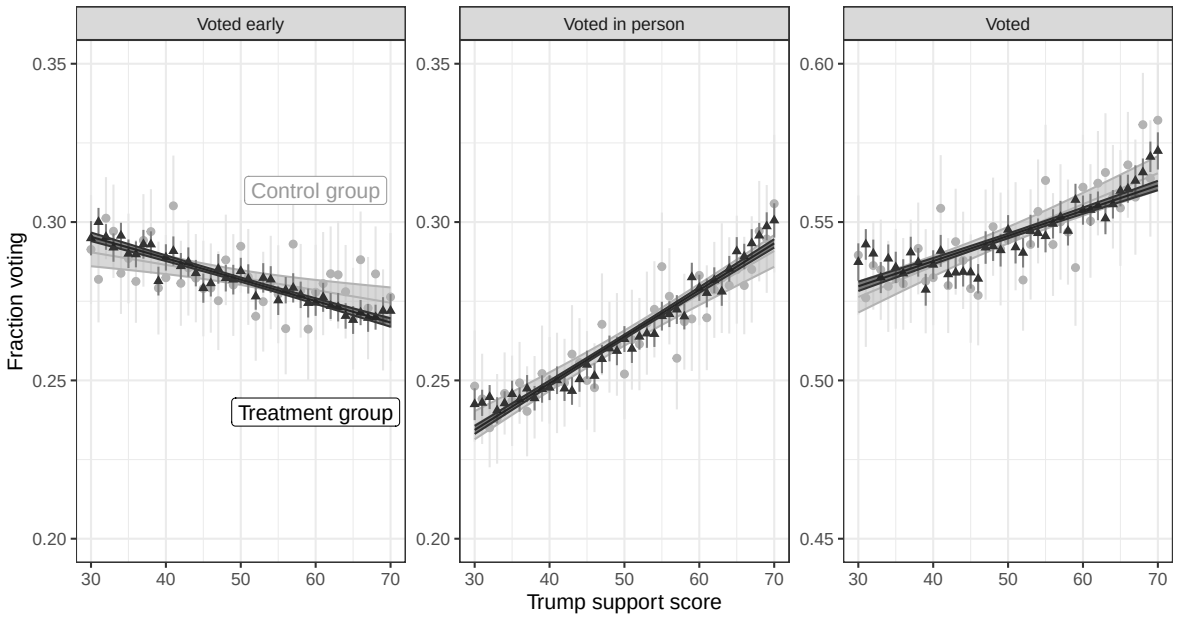


Figure 3: Figure 3 as the rewrite draws it. The two group labels come from the deposited script; the published figure does not carry them.

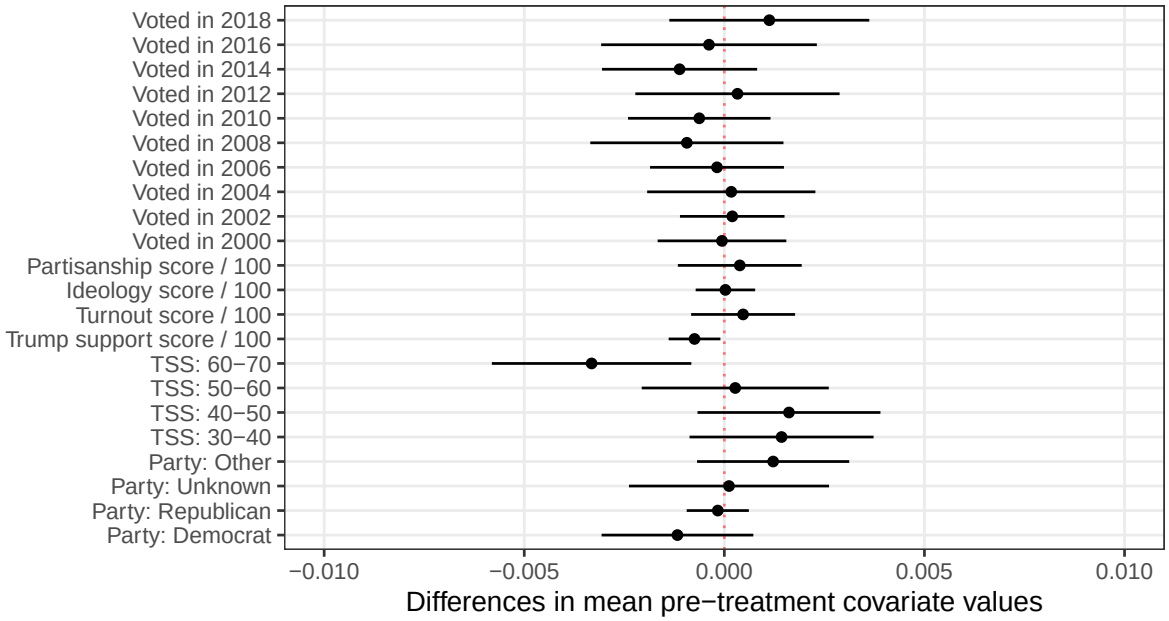


Figure 4: Figure 5 as the rewrite draws it, with the deposited script's own covariate labels.



Figure 5: Supplementary Figure 2 as the rewrite draws it.

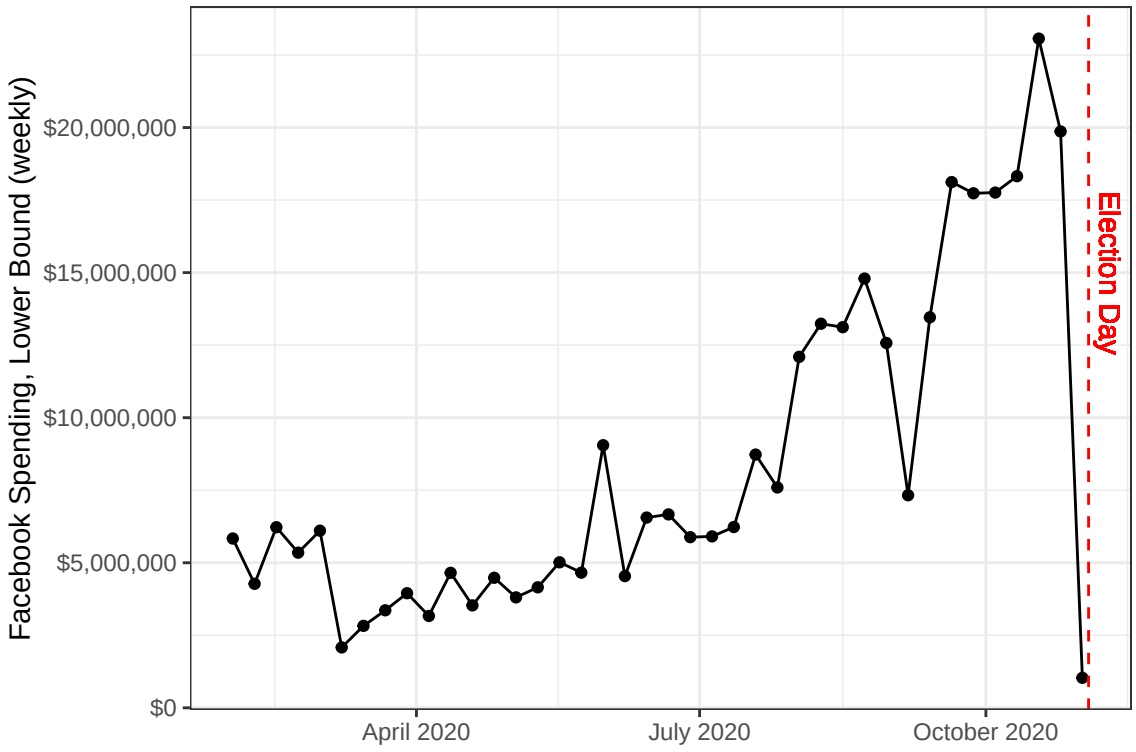


Figure 6: Supplementary Figure 3 as the rewrite draws it.

9 Rewrite verification

`run_all.R` was run to completion twice from clean sessions and the whole tree diffed. Every CSV and TeX file is byte-identical between runs. The figure PDFs differ, because a PDF records the time it was written.

`original/` is verified twice per run. `download_original.R` is sourced first, which checks every file against the MD5 Dataverse serves for `?format=original` and against its deposited byte size, and then refuses to continue if `original/` holds any file the manifest does not list, dotfiles included. It is sourced again as the last step, because the first pass proves only that the archive was intact when the run began. All 16 files match, and 0 carry a published checksum that disagrees with the bytes Dataverse serves. Two of the sixteen are ingested as tabular data and are served under a `.tab` name unless the original format is requested; the manifest records both names. Checksums for files this repository writes are not quoted anywhere: a PDF records its write time, so its hash changes on every run.

10 R environment

```
Warning in styling_latex_scale(out, table_info, "down"): Longtable cannot be resized.
```

component	version
R	4.6.2
tidyverse	2.0.0
estimatr	2.0.1
car	3.1.5
lmtest	0.9.40
ggplot2	4.0.3

The article reports R 4.1.1 with tidyverse 1.3.1, estimatr 0.30.6 and car 3.1.0. Nothing in this analysis draws at random: no script calls `sample()`, `rnorm()` or a bootstrap, so the results are bit-identical at any seed and the sampler change in R 3.6 is not in play. The one place where composing two packages could have moved a published number is the omnibus balance test, which calls `lmtest::waldtest()` on two `estimatr::lm_robust` objects. Current `estimatr` exports a `vcov` method that the version of the day did not, which is the mechanism that silently changed an entire published table in another archive in this programme. Here it does not bite: the test returns 1.0266 on 18 and 1,999,246 degrees of freedom, which is the published statistic exactly.